

BSc (Hons) Computing

A Comparative Evaluation of Time Series Forecasting Models for Glucose Prediction in Type 1 Diabetes Mellitus

CLD6001

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Abstract

This thesis conducts a rigorous comparative evaluation of diverse time series forecasting models for short-term glucose prediction in Type 1 Diabetes (T1D) management. The UVA/Padova T1D simulator was used (28 virtual patients, 30 days of CGM, insulin, and carbohydrate data), the study implemented classical statistical methods (ARIMA, Exponential Smoothing), traditional machine learning, and advanced deep learning architectures including LSTM, N-BEATS, Temporal Fusion Transformer (TFT), and Transformer. The statistical performance was measured by using the Mean Absolute Error (MAE), and the Root Mean Squared Error (RMSE), while the clinical safety via Clarke Error Grid (CEG) analysis. Generalizability was rigorously tested through backtesting, 10-fold expanding window cross-validation, and evaluation on entirely unseen patient data.

The Transformer emerged as the most potent non-linear model, achieving an MAE of $10.18 \frac{mg}{dL}$ and RMSE of $14.13 \frac{mg}{dL}$ on unseen patients, with 98.1% of predictions falling within clinically safe CEG Zones A/B. The ARIMA model proved to be the most effective linear model (cross-validation MAE: 12.59 mg/dL, RMSE: 15.64 mg/dL), demonstrating strong baseline performance. The TFT, despite its sophistication, underperformed due to overfitting. This research establishes a robust benchmark, underscoring the significant potential of deep learning, particularly Transformer architectures, for developing accurate and clinically reliable glucose prediction systems, while also highlighting the critical importance of comprehensive evaluation methodologies.

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1 Introduction

Diabetes mellitus is a pervasive chronic metabolic disorder that affects hundreds of millions of individuals worldwide, and imposes a substantial burden on healthcare systems and patients quality of life. The condition is characterized by elevated blood glucose level (hyperglycemia) caused by defects in insulin secretion, insulin action, or both. Type 1 diabetes (T1D) in particular is an autoimmune disease where the body's immune system mistakenly attacks its own cells. In particular it attacks insulin-producing beta cells in the pancreas. Once these cells are destroyed, the body can no longer produce insulin. Consequently, individuals with T1D produce little to no insulin, a hormone that enables cells to absorb glucose from the bloodstream for energy. Without exogenous insulin, type 1 diabetes is life threatening.

Managing T1D is an intricate and relentless balancing act, where individuals must meticulously monitor their glucose levels and administer the appropriate doses of insulin, either by insulin injection or insulin pump. Furthermore, they must also count carbohydrate intake, and also consider the impact of various other glycemic disruptors, such as stress, physical activity, illness, hormonal changes. Self-management is very taxing for individuals with T1D, often leading to significant psychological distress, reducing an individual's quality of life. The primary goal of T1D management is to maintain blood glucose level within a specified target range, this typically is between $70\text{-}180 \frac{\text{mg}}{\text{dL}}$ or $4.4\text{-}10 \frac{\text{mmol}}{\text{L}}$ to prevent acute and chronic complications. Acute complications include dangerously low blood glucose, called hypoglycemia, which leads to confusion, seizures, loss of consciousness, and if left untreated even death. Another complication of T1D is ketoacidosis (DKA). DKA happens when the body does not have enough insulin, so the body starts breaking down fats instead of sugar. This breakdown of fats produces ketones, which are acids. When too many ketones build up in the blood, it makes the blood pH drop, reducing the normal Arterial Blood Gas (ABG) values from the normal pH range of (7.35-7.45) to below 7.30.

On the other hand, chronic hyperglycemia significantly increases the risk of long-term microvascular and macrovascular complications.

The emergence of Continuous Glucose Monitoring (CGM) systems has completely revolutionized diabetes care, moving from Self-Monitoring of Blood Glucose (SMBG) using finger sticks blood glucose measurements, to automated glucose monitoring. Modern CGM systems also include alarms for high and low glucose levels, giving the opportunity to the user to take corrective actions in time. The data generated by CGM systems is dense enough to be considered an exceptional candidate for applying time series forecasting techniques.

Accurate and reliable short-term glucose prediction is a cornerstone of modern and future diabetes management. By managing to forecast any glycemic excursions individuals with T1D can transition from a reactive management approach to a proactive one. For example, a prediction of hypoglycemia can tell the patient to consume some carbohydrates before any symptoms appear, while a forecast of postprandial hyperglycemia can guide the user to adjust insulin doses for that specific instance. Furthermore, glucose prediction algorithms are integral components of advanced diabetes technologies, particularly Automated Insulin Delivery (AID) systems, more commonly called “artificial pancreas” systems. These systems make use of the CGM data available and a control algorithm to automatically adjust insulin delivery using an insulin pump, in order to keep glucose level at an euglycemic range with minimal to none human intervention. The accuracy and reliability of the glucose prediction directly impact the performance and safety of these AID systems.

However, modeling and predicting blood glucose dynamics is a very complex task due to the inherent complexity of human physiology, and interacting variables. Insulin dynamics play a significant role, this includes the type of insulin used, basal or bolus, the absorption rate from subcutaneous tissue, and the insulin sensitivity of each individual.

Carbohydrate metabolism also plays a major role, this is influenced by the quality and the type of consumed carbohydrates consumed, the digestion and absorption rates, and

the presence of other macronutrients like protein and fats can delay glucose absorption.

Physical activity adds another layer of complexity to glucose prediction, as the type of exercise, the duration and the intensity can lead to a drop of glucose level and increased risk of hypoglycemia to meet the energy demands required by the muscles.

Circadian rhythms and hormonal fluctuations also contribute, with natural diurnal variations in hormone levels can affect insulin sensitivity.

Inter-individual, and intra-individual variability is another contributing factor, because each individual responds differently to insulin, food and exercise. Even within the same person, some responses can have very different effects from one day to another.

These complexities in glucose are the reason why glucose behaves non-linearly and as time series they are typically non-stationary, exhibiting multi-scale temporal dependencies, and often influenced by exogenous covariates. While CGM technology provides all this data, it is not without its own set of challenges, sensor noise, missing data values, and calibration drifts are a real issue that can greatly hinders the model's ability to make high quality predictions.

This thesis embarks on a rigorous comparative evaluation of different time series forecasting models for short-term glucose level prediction. The study makes use of data generated by the UVA/Padova Type 1 Diabetes Simulator, an FDA-accepted tool for in silico testing of diabetes technologies. The spectrum of models implemented ranges from classical statistical methods like ARIMA and exponential smoothing, to traditional machine learning methods, and culminating in deep learning architectures such as LSTMs, N-BEATS, and a Transformer architecture.

The main objective for this research are:

- To implement and thoroughly evaluate all the selects models on a standardized, high-fidelity simulation dataset with a cohort of 28 virtual patients of varying ages with T1D, each with 30 days of CGM, insulin, and carbohydrate intake.
- To asses model performance not only based on standard statistical error metrics

like Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE), but also in terms of their clinical safety and relevance using Clarke Error Grid (CEG)

- To rigorously investigate the generalizability of the model through backtesting and an expanding window cross-validation strategy, including testing on unseen patient data to mimic real-world scenarios.
- To identify the most promising model architectures that can manage to balance predictive accuracy, computational efficiency, and clinical safety.

This thesis is structured to systematically address all of the aforementioned objectives.

2 Literature Review

2.1 Time Series Forecasting in Healthcare

Glucose prediction presents itself as a time series forecasting problem. Historical glucose measurements alongside relevant covariates such as: insulin delivery (bolus and basal) and carbohydrate intake are used to forecast glucose levels in the future. Time series forecasting is the process of using historical data to predict future values of a variable. This process entails modeling temporal dependencies and patterns such as: seasonality, trends, and autocorrelation using different types of statistical, machine learning, or deep learning methods (Box, 2016).

For blood sugar levels in particular, it involves analyzing data over time and examining different aspects of it. Blood glucose levels as a time series are typically non-stationary, this means that they have varying statistical properties due to some physiological factors such as circadian rhythms, meal timing, physical activity, and emotional states (Wang, Tong, and M. Yu, 2020). Furthermore, glucose regulation in general represents a complex physiological system with multiple interacting factors that are still being actively researched

like insulin sensitivity, carbohydrate metabolism, stress hormones, and physical activity (Gangopadhyay and Mukherjee, 2023; Koufakis et al., 2019).

The development of Continuous Glucose Monitoring (CGM) systems has revolutionized glucose time series forecasting by providing not only high-quality data but also high-frequency measurements at regular intervals (typically every 3–15 minutes) throughout the day (Martinsson et al., 2019). These recent technologies have enabled researchers to pursue more sophisticated forecasting methods able to capture both micro-fluctuations and macro-patterns in glucose dynamics at the cost of higher data volume (Herrero et al., 2024).

The large volume of longitudinal data has aided more data-intensive modeling approaches that were previously unfeasible with Self-Monitoring of Blood Glucose (SMBG) measurements (Wang, Tong, and M. Yu, 2020).

time series forecasting in healthcare, through time, has evolved from univariate methods, which consider only the target variable, to multivariate methods that incorporate relevant covariates. Glucose prediction in particular typically incorporates carbohydrate intake, insulin dosing, physical activity, and other physiological markers that have led to some substantial improvements in prediction accuracy (Doorn et al., 2021). Predicting blood glucose levels one hour into the future can help patients who use an automated insulin delivery system anticipate and prevent hypoglycemic and hyperglycemic events by adjusting insulin doses or carbohydrate intake in advance. This is essential for patients with type 1 diabetes, as timely interventions can prevent dangerous blood glucose excursions (Acuña and Aparicio, 2023; Kitchainukoon and Sae-Bae, 2023).

2.1.1 Challenges in Glucose Time Series Analysis

Glucose prediction poses distinct challenges compared to other healthcare time series problems. The inherent variability in glucose dynamics, influenced by a plethora of exogenous covariates, including meal consumption, insulin absorption rate, physical activity

and intensity, and physiological stress—creates a complex prediction landscape. Furthermore, inter-individual and intra-individual variability hinders the development of more generalized models. Additionally, carbohydrate intake and insulin doses can produce different glycemic responses across different patients and even within the same patient under different circumstances (Sevil et al., 2021).

Another challenge specific to CGM systems is data availability and data quality, including sensor noise, missing values, and calibration errors. These issues require robust preprocessing methodologies to guarantee that the model receives high-quality and reliable inputs (Kozinetz et al., 2024).

In addition, the multi-scale temporal dynamics of glucose regulation—such as insulin action time (minutes), meal digestion (hours), and circadian rhythm (days) require models that are able to recognize temporal patterns at multiple time scales simultaneously (Doorn et al., 2021).

2.2 Statistical Forecasting Methods for Glucose Prediction

2.2.1 ARIMA

Autoregressive Integrated Moving Average (ARIMA) models have been extensively applied to glucose prediction due to their capacity to encapsulate temporal dependencies in time series data. These models decompose the time series into autoregressive (AR) components, representing dependencies on past observations; integrated (I) components, to address non-stationarity; and moving average (MA) components, modeling forecast errors (Box, 2016). (Prendin et al., 2023) implemented an *ARIMA* model with an *RMSE* of $23.78 \frac{mg}{dL}$. The primary advantages of ARIMA approaches include their interpretability, low computational requirements, and established statistical frameworks. However, they face limitations in capturing complex non-linear relationships and in incorporating exogenous variables, such as bolus and basal insulin doses and meal information (Melina et al.,

2024).

Recently ARIMA models have been extended, like for example, Seasonal ARIMA (SARIMA) which in term have improved their ability to perform glucose level prediction. SARIMA models are able to capture the daily seasonality in glucose fluctuations. Some other ARIMA models like ARIMAX models, which incorporate exogenous variables, hence the X, have demonstrated better performance by including insulin and meal data (Rancati et al., 2024).

2.2.2 Exponential Smoothing

Another class of statistical forecasting methods is Exponential smoothing. These models calculate the weighted averages of past observations, but the greater the temporal difference the less their weight becomes. The weight of each time-step exponentially decreases over time, this process gives a higher weight in the weighted average to newer time-steps (Brockwell and Davis, 2016).

The main advantages of exponential smoothing are its simplicity, their computational efficiency, and their robustness to some data irregularities. However, their performance quickly falls off when trying to predict longer prediction horizons (PH) or when exogenous factors, such as meals or exercise, drastically alter the behavior of the time series (Facchinetti et al., 2015).

Recent research has managed to make some improvements and refinements to exponential smoothing models for glucose prediction. Triple exponential smoothing, also known as the Holt-Winters method, is a special type of exponential smoothing model, which is able to capture level, trend, and seasonality. This version of exponential smoothing has managed to achieve some moderate improvements over simpler model variations in different fields (Christioko, Khoirudin, and Daru, 2023).

These probabilistic forecasts enable risk-aware decision-making by providing not only point predictions but also confidence bounds for future glucose values.

2.2.3 Linear Regression

Linear regression models create a direct relationship between the target variables and their predictors, making them conceptually straightforward for blood glucose level prediction. A comparative study using the in-silico UVA/Padova type 1 diabetes simulator implemented regularized linear regression models, achieving a MAE of $9.2 \frac{mg}{dL}$ on a 30-minute prediction horizon. This work demonstrated that linear regression models can be substantially improved through careful feature extraction (Mirshekarian et al., 2019).

While regression approaches offer a high degree of interpretability and computational efficiency, they struggle to capture the complex dynamics of blood glucose. Nevertheless, they remain valuable as baseline models.

Advanced regression techniques have improved the performance of these linear models. Ridge regression and LASSO regression have been widely used to address multicollinearity and perform feature selection in various regression problems, improving model stability and interpretability. Elastic Net, which combines the two methods, has been used to balance sparsity and coefficient stability, as supported by recent research comparing these methods across different domains (Herawati et al., 2024; Usman, Doguwa, and Alhaji, 2022).

2.2.4 Kalman Filters

Kalman filters are an important class of statistical forecasting methods for glucose prediction, bridging the gap between physiological glucose-insulin dynamics (e.g., interstitial fluid delay models) and sensor data. These recursive state estimators are well-suited for handling the inherent noise from CGM sensors while simultaneously incorporating known dynamics of glucose-insulin interactions (Halvorsen et al., 2022; Kim and Jung, 2024).

The Extended Kalman Filter (EKF) has been used for glucose forecasting thanks to its ability to address the nonlinear dynamics of glucose-insulin interactions and filtering noisy CGM data. Specifically, EKFs are able to linearize nonlinear glucose-insulin models

by creating a linear approximation around the current state estimate. This allows the system to continuously update the predictions and recursively correct errors as more data becomes available (Halvorsen et al., 2022). This approach enables real-time state and parameter estimation, fault detection, and meal detection for artificial pancreas systems (closed-looped systems). One study has also implemented EKF on Android smartphones to demonstrate their practical applicability for CGM and glucose control (Mahmoudi et al., 2019).

What makes Kalman filters particularly powerful for blood glucose prediction is their ability to incorporate physiological knowledge through state transition models, robustly handle measurement noise, and perform real-time updates—capabilities that are essential for a closed-loop insulin delivery system (Griva, Outbib, and Basualdo, 2022).

2.3 Machine Learning Approaches to Glucose Prediction

2.3.1 Traditional Machine Learning Methods

The limitations of statistical approaches have motivated researchers to adopt machine learning techniques for glucose prediction. Random forests and gradient boosting trees have demonstrated higher-quality forecasts, primarily due to their capacity to encapsulate non-linear relationships without requiring explicit specification of the functional form (Kozinetz et al., 2024).

Traditional machine learning models offer several advantages over classical statistical forecasting methods, notably the ability to model complex non-linear relationships and incorporate relevant covariates. However, they require feature engineering and may not fully exploit the sequential nature of blood glucose time series (Kurt et al., 2023).

Recent advancements in gradient boosting methods have further improved prediction accuracy. An XGBoost implementation for glucose level prediction outperformed traditional random forests by leveraging regularization techniques that prevent overfitting

(Praveen et al., 2023; Yilin Wang, 2023).

2.3.2 Support Vector Regression

Support Vector Regression (SVR) has emerged as an effective technique for glucose prediction, mainly due to its ability to handle non-linear data through kernel transformations while maintaining good generalization properties (S, 2019).

Some studies have also found that radial basis function (RBF) kernels outperform linear and polynomial kernels in glucose prediction tasks, successfully modeling more complex relationships (S, 2019).

The main advantages of SVR are its robustness to outliers and its ability to handle high-dimensional data. However, its primary limitations include computational complexity for large datasets and challenges with hyperparameter tuning, which impact the feasibility of clinical implementation (Syafudin et al., 2022).

2.4 Advanced Deep Learning Architectures

2.4.1 Long Short-Term Memory Networks

Long Short-Term Memory (LSTM) networks are a specialized type of recurrent neural networks. Their rise to popularity in recent years stems from the fact that they are particularly effective at modeling sequential data like time series.

Unlike traditional feedforward networks (FFNs), LSTMs use memory cells and gating mechanisms that selectively keep (“remember”) or discard (“forget”) information. This makes them particularly useful for capturing the complex temporal dynamics that are part of glucose regulation.

The main advantages of LSTM networks for glucose prediction is their ability to model the necessary temporal patterns without extensive feature engineering. Furthermore, their capacity to handle non-linear data, and their flexibility in incorporating multivariate inputs

makes them the ideal candidate for glucose level prediction (Shuvo et al., 2024; Xiong, Ren, and Liu, 2024). However, these benefits come at some costs, mainly: the increased computational requirements, the reduced interpretability, and also a higher risk of overfitting compared to simpler models (Dylag, 2023).

Architectural improvements have further improved traditional LSTM performance in a variety of different fields (e.g. electrical engineering). Bidirectional LSTMs have improved their accuracy compared to a traditional LSTM by processing glucose sequences both forward and backward, enabling the model to capture dependencies that might be missed by a unidirectional approach (Alizadegan et al., 2024). Attention mechanisms combined with LSTMs have shown enhanced prediction performance across various time series domains by focusing on relevant historical patterns (Wu, 2023).

Transfer learning approaches have also been implemented to address data scarcity, pre-training models on population-level data and subsequently fine-tuning them on individual data.

2.4.2 Convolutional Neural Networks

Convolutional Neural Networks (CNNs) are a network architecture traditionally associated with image processing, but in recent years they have been successfully adapted for glucose time series forecasting by treating glucose data as one-dimensional temporal signals. CNNs can automatically extract features across different temporal scales through convolutional layers (D’Antoni et al., 2022; Xiong, Ren, and Liu, 2024).

Dilated causal convolutions have also proven particularly useful for blood glucose prediction by expanding the receptive field without increasing the number of parameters. This, in turn, allows the model to capture both short-term and long-term dependencies (Cenerini et al., 2024).

Hybrid architectures that combine convolutional layers for feature extraction and recurrent layers (LSTM, GRU) for temporal modeling have achieved state-of-the-art perfor-

mance in glucose level prediction, reducing error compared to standalone CNNs (Xiong, Ren, and Liu, 2024).

2.4.3 Transformer Models

The transformer architecture was originally developed for natural language processing (NLP), specifically introduced to address language translation tasks (Vaswani et al., 2023). Their strong performance in NLP motivated researchers to adapt transformers for image and video tasks, as well as for time series forecasting, where capturing temporal dependencies is essential.

These models employ the self-attention mechanism, which models relationships between all time points in a sequence, offering, in some cases, better performance than LSTMs (Lv et al., 2024; Zhou et al., 2021).

(K. Li et al., 2020) applied transformer models to glucose prediction using continuous glucose monitoring (CGM) data, showing that for prediction horizons of 60 to 120 minutes, transformers outperformed LSTM architectures. Their analysis revealed that the self-attention mechanism enabled transformers to better capture both short-term and long-term dependencies, including patterns such as circadian rhythms.

Transformers offer several advantages for glucose prediction, including parallel processing of data, which in turn reduces training times; superior modeling of complex temporal dependencies through the self-attention mechanism; and scalability to incorporate multivariate inputs for improved forecasts (Hakim et al., 2024). However, transformers generally require substantially larger data volumes to achieve optimal performance compared to simpler models (D. Liang et al., 2024).

Recent advancements in transformer architectures include time-aware attention mechanisms that explicitly model the temporal distances between data points, improving performance in cases of irregular sampling or missing data (Cheng et al., 2024).

Patient-specific adaptation techniques have also been developed to fine-tune pre-

trained transformers toward individual patient patterns. These personalization strategies have reduced the gap between population-level models and individualized forecasts, making transformers increasingly practical for clinical applications (S. Li et al., 2024).

2.4.4 N-BEATS

The Neural Basis Expansion Analysis for Interpretable Time Series Forecasting (N-BEATS) represents a specialized deep learning architecture designed specifically for time series prediction. Unlike general-purpose neural networks, N-BEATS incorporates domain knowledge about time series decomposition, modeling trend and seasonal components explicitly while retaining the flexibility of deep learning approaches (Oreshkin et al., 2020).

This ability to simultaneously consider both recent measurements and historically similar contexts explains the model's superior performance for extended prediction horizons, where traditional recurrent architectures struggle due to diminishing influence from distant but relevant past measurements.

Applications of N-BEATS to glucose prediction have demonstrated particular effectiveness in capturing the multi-scale temporal patterns characteristic of glycemic dynamics. The hierarchical nature of the architecture's forecast blocks enables it to model both short-term effects, such as the immediate impact of insulin and carbohydrate intake, alongside longer-term physiological rhythms (Zhu, Chen, Kuang, et al., 2023).

2.4.5 Temporal Fusion Transformer

The Temporal Fusion Transformer (TFT) represents a recent advancement specifically designed for multi-horizon time series forecasting with mixed data types (Lim et al., 2020). This architecture combines the use of recurrent neural networks for local processing with attention mechanisms for capturing long-term dependencies, while also enabling the modeling of different types of covariates (static, known future, and unknown future covariates).

The TFT architecture offers several advantages for glucose prediction, including explicit covariate selection for improved interpretability and uncertainty quantification through the use of quantile forecasting (Zhu, Chen, Kuang, et al., 2023).

These characteristics make TFT particularly desirable for clinical applications, where model interpretability and reliability are paramount considerations.

2.4.6 Deep Reinforcement Learning

Although deep reinforcement learning (DRL) approaches are not strictly forecasting models, they frame glucose management as a sequential decision-making problem, where agents learn optimal insulin delivery policies to maintain glucose levels within the target range (Viroonluecha, Egea-Lopez, and Santa, 2022; Zhu, K. Li, and Georgiou, 2023).

Actor-critic architectures, such as Deep Deterministic Policy Gradient (DDPG) methods, have demonstrated effectiveness in UVA/Padova simulations, showing significant improvements over traditional control algorithms (Zhu, K. Li, and Georgiou, 2023).

The integration of uncertainty-aware glucose predictions and risk-sensitive objectives, which penalize dangerous glycemic excursions, has enhanced the robustness of these approaches. Risk-sensitive reinforcement learning explicitly penalizes actions that could lead to unsafe glycemic levels (X. Yu et al., 2025).

While clinical validation remains limited, initial studies on hybrid closed-loop systems incorporating DRL have shown promising results regarding safety and efficacy compared to fully autonomous control systems (Zhao et al., 2025; Zhu, K. Li, and Georgiou, 2023).

The ability of DRL models to adapt to changing patient conditions through online learning represents a significant advantage over static models, particularly for patients with variable insulin sensitivity or irregular lifestyles (X. Yu et al., 2025).

2.5 Hybrid and Ensemble Methods

2.5.1 Ensemble Methods and Model Stacking

Ensemble methods have consistently shown improved performance compared to individual models by taking advantage of their complementary strengths and, at the same time, mitigating weaknesses (Karunarathna and Z. Liang, 2024; Saiti, Macaš, Lhotská, et al., 2020). When used for forecasting blood glucose level for type 1 diabetes, ensemble methods have been shown to provide superior prediction accuracy, as they accommodate multiple model outputs to account for variability in physiological responses (Saiti, Macaš, Lhotská, et al., 2020).

Weighted averaging ensembles dynamically which scale contributions based on recent performance, have achieved high accuracy across varying glycemic states and patient activity (Karunarathna and Z. Liang, 2024). More sophisticated ensembles that utilize gradient boosting meta-learners have also shown better performance in more complex prediction tasks (Boateng and Yang, 2023).

3 Methodology

3.1 Dataset Description

This project's data was created in a simulation program written in Python by (Xie et al., 2025). The simulation uses the UVA/Padova Type 1 Diabetes Simulator. This simulator has been approved by the U.S. Food and Drug Administration (FDA) for preclinical testing of glucose control algorithms. The simulation provides high-fidelity metabolic data that closely resemble real-world glucose dynamics in patients with Type 1 diabetes (Man et al., 2014).

The dataset consists of 28 patients: 10 adults, 10 adolescents, and 8 children. Each patient was assigned a random scenario, with a seed of 42 used for reproducibility. All

readings were obtained using a simulated Guardian CGM sensor, with a random noise function initialized using the same seed of 42. Insulin delivery was simulated using a Cozmo insulin pump with a standard basal-bolus controller. All the patient data had a 5-minute time step between CGM reading over a continuous 30-day period, resulting in 8,640 data points per patient. The dataset contains no missing values, providing the best quality data possible, ideal for model development and evaluation.

The simulation returned these four variables per patient:

- Continuous Glucose Monitoring (CGM): Glucose levels measured in ($\frac{mg}{dL}$).
- Carbohydrate Intake (CHO): Measured in grams (g).
- Bolus Insulin: Measured in units (U).
- Basal Insulin: Measured in units per hour ($\frac{U}{hr}$).

3.2 Data Preprocessing

3.2.1 One-Compartment Open Model

A one compartment model can be used to the distribution of the bolus insulin in the system as well as its elimination is a simplified way (Ahmed, 2015).

The equation describing the model is:

$$C(t) = Ae^{-k_{el}t} \quad (1)$$

where $C(t)$ represents the drug concentration at a specific time point after the reference time, starting from the initial concentration. A refers to the initial concentration, which is equal to the dose of bolus insulin measured in units (U). This insulin typically begins to act within 5 to 15 minutes and remains active for approximately 3 to 4 hours (Saiti, Macaš, Štechová, et al., 2018).

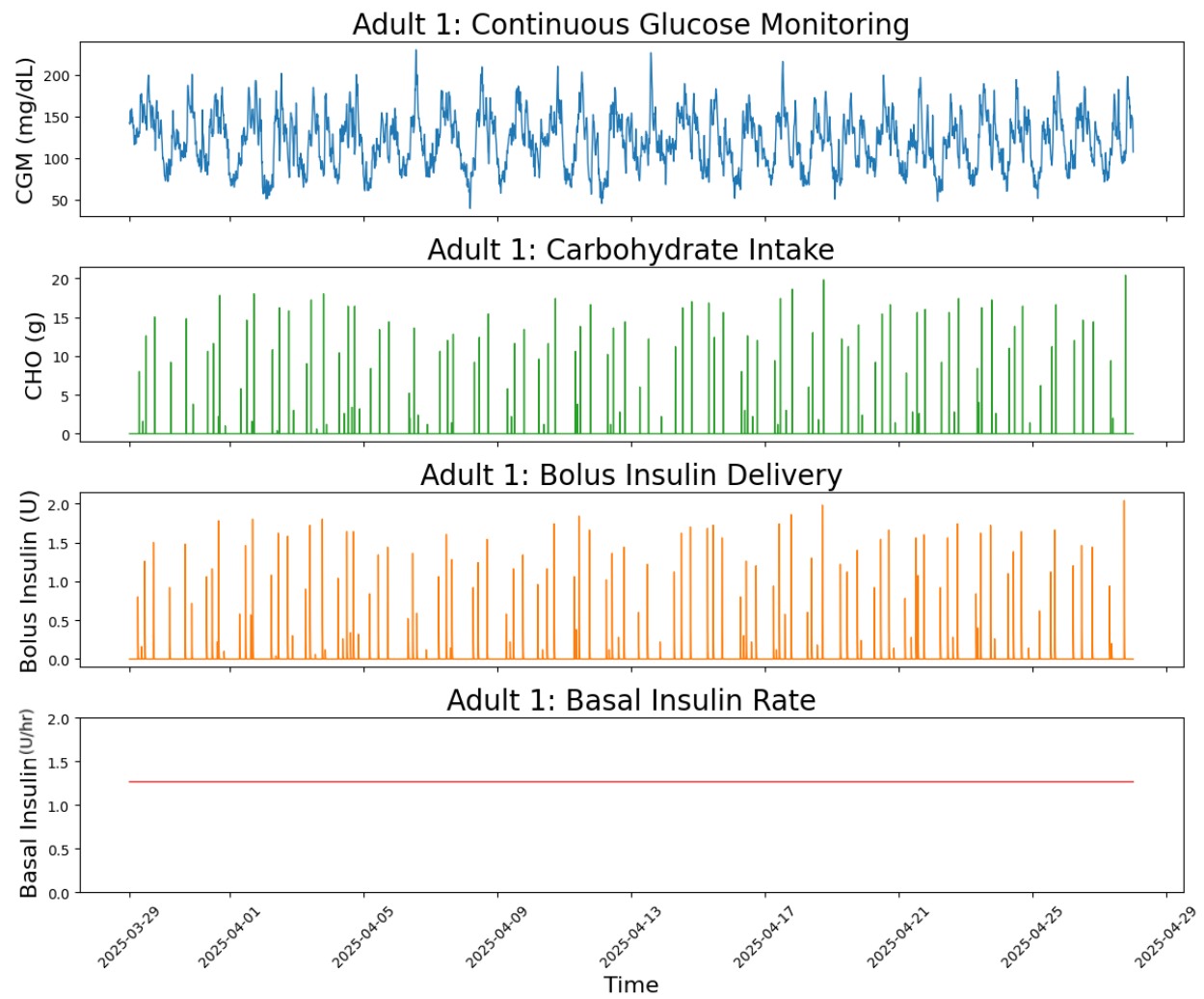


Figure 1: Input data: adult 001



Figure 2: One-compartment open model (Ahmed, 2015)

The elimination rate k_{el} describes how quickly the drug is eliminated from the body and is calculated using the following formula:

$$k_{el} = \frac{\ln(2)}{t_{1/2}} \quad (2)$$

where $t_{1/2}$ is the insulin's half-life. In this case, $t_{1/2}$ is 5 minutes, resulting in an elimination rate of $k_{el} = 0.138 \text{ min}^{-1}$ (Saiti, Macaš, Štechová, et al., 2018).

3.2.2 Feature Scaling

To enhance model convergence and performance during training, two types of scalers were applied to the data. A *Min-Max* scaler was applied to CGM values to normalize the range to $[0, 1]$ using the following formula:

$$X_{scaled} = \frac{X - X_{min}}{X_{max} - X_{min}} \quad (3)$$

While for the covariates, carbohydrate intake, bolus insulin, and basal insulin, a standard scaler was used to achieve zero mean and variance of one using this formula:

$$X_{scaled} = \frac{X - \mu}{\sigma} \quad (4)$$

where μ is the mean and σ is the standard deviation.

3.2.3 Train-Test Split

The dataset was initially split using a 90-10 ratio, with 90% of the data (approximately 27 days) used for model training, and the remaining 10% (approximately 3 days) reserved for validation.

For the two best-performing models identified in the initial evaluation (ARIMA and vanilla Transformer), additional cross-validation was performed using a sliding window

approach to ensure a more robust validation assessment.

3.3 Prediction Models

Eight distinct models with varying architectures were implemented and evaluated. Each model offers unique advantages in capturing temporal dependencies, temporal patterns, and interpretability in glucose time-series data.

3.3.1 ARIMA: Autoregressive Integrated Moving Average

ARIMA is a classical statistical method that combines autoregressive (AR), differencing (I), and moving average (MA) components to model time-series data (Box, 2016).

The general ARIMA(p, d, q) model is (Box, 2016):

$$\phi(B)(1 - B)^d y_t = \theta(B)\epsilon_t \quad (5)$$

The model is defined by three parameters:

- p : which is the number of lag observations.
- d : which is the degree of differencing.
- q : which is the size of the moving average window.

where:

- B is the backshift operator, defined as $By_t = y_{t-1}$.
- $\phi(B) = 1 - \phi_1 B - \dots - \phi_p B^p$ is the AR polynomial.
- $\theta(B) = 1 + \theta_1 B + \dots + \theta_q B^q$ is the MA polynomial.
- d is the order of differencing (used to remove trends).
- ϵ_t is white noise.

Usually, the parameters ϕ_i and θ_j are calculated by applying maximum likelihood or least squares on a stationary differenced series.

The implementation used an ARIMA(12, 1, 1) configuration with a prediction horizon of 60 minutes (12 time steps), and rolling window forecasts were applied during back-testing.

3.3.2 Exponential Smoothing

Exponential smoothing methods assign exponentially decreasing weights to observations as they become further away from the current time step T . This enables the model to take past patterns into consideration while giving more weight to more recent observations.

When a time-series does not exhibit any patterns or seasonality, simple exponential smoothing is applied (Brown and Meyer, 1961). The smoothed forecast value at time t is:

$$\hat{y}_{t+1} = \alpha y_t + (1 - \alpha)\hat{y}_t \quad (6)$$

Where:

- α is the smoothing parameter $0 < \alpha < 1$.
- $\hat{y}$ is the forecast for period t .
- Requisitely expands to: $\hat{y}_{t+1} = \alpha y_t + (1 - \alpha)y_{t-1} + \alpha(1 - \alpha)^2 y_{t-2} + \dots$

Making more recent time steps have a bigger impact on the prediction.

When a time-series exhibits both trend and seasonality, a more advanced exponential smoothing model is used. The Holt-Winters Method takes into consideration both trend and seasonality (Box, 2016).

Level Update:

$$l_t = \alpha(y_t - s_{t-m}) + (1 - \alpha)(l_{t-1} + b_{t-1}) \quad (7)$$

Trend Update:

$$b_t = \beta(l_t - l_{t-1}) + (1 - \beta)b_{t-1} \quad (8)$$

Seasonal Update:

$$s_t = \gamma(y_t - l_{t-1} - b_{t-1}) + (1 - \gamma)s_{t-m} \quad (9)$$

Forecast equation:

$$\hat{y}_{t+h} = l_t + hb_t + s_{t-m+(h \bmod m)} \quad (10)$$

Where:

- γ is the seasonal smoothing parameter.
- m is the seasonal period.

3.3.3 Linear Regression

Linear regression is a foundational statistical method used to model relationships between a dependent variable and one or more independent variables. In the context of time-series forecasting, the goal is to predict future values of a series y_t using information from past observations (Montgomery, Peck, and Vining, 2021).

A basic linear regression model is:

$$y_t = \beta_0 + \beta_1 x_{1,t} + \beta_2 x_{2,t} + \cdots + \beta_p x_{p,t} + \epsilon_t \quad (11)$$

where:

- y_t is the observed value of the target variable at time t ,
- $x_{j,t}$ is the value of the j -th predictor at time t ,
- β_0 is the intercept,
- β_j are the regression coefficients representing the influence of predictor j ,
- ϵ_t is the random error term at time t , assumed to follow a normal distribution with zero mean and constant variance ($\epsilon_t \sim N(0, \sigma^2)$).

The model is estimated by minimizing the sum of squared residuals, yielding ordinary least squares (OLS) estimates for the coefficients β_j :

$$\min_{\beta} \sum_{t=1}^n \left(y_t - \beta_0 - \sum_{j=1}^p \beta_j x_{j,t} \right)^2 \quad (12)$$

In time-series applications, a common strategy is to use lagged values of the dependent variable as predictors, essentially creating an autoregressive regression:

$$y_t = \beta_0 + \beta_1 y_{t-1} + \beta_2 y_{t-2} + \cdots + \beta_p y_{t-p} + \epsilon_t \quad (13)$$

To make forecasts with the trained model, known or forecasted predictor values are plugged into the model to generate future estimates. For example, a one-step-ahead forecast is given by:

$$\hat{y}_{t+1} = \hat{\beta}_0 + \hat{\beta}_1 y_t + \hat{\beta}_2 y_{t-1} + \cdots + \hat{\beta}_p y_{t-p} \quad (14)$$

For multi-step-ahead forecasts, the model uses an iterative approach, where previously forecasted values are fed back as inputs.

To ensure a fair comparison with other deep learning models, the implementation used 72 time steps (6 hours) as input and predicted 12 time steps (1 hour) at a time.

3.3.4 LSTM (Long Short-Term Memory)

LSTM is a specialized type of Recurrent Neural Network (RNN) architecture specifically designed to mitigate the vanishing/exploding gradient problem that traditional RNNs suffer from. This characteristic makes LSTMs well equipped for learning long-term dependencies in sequential data (Hochreiter and Schmidhuber, 1997).

The key innovation of LSTMs is the use of special structures called memory cells, which include three types of gates:

- The forget gate, which decides what information to discard from the cell state.
- The input gate, which decides what new information to store in the cell state.
- The output gate, which decides what part of the cell state to output.

The core equations governing the LSTM cell are as follows (Graves, Jaitly, and Mohamed, 2013):

Forget gate:

$$\mathbf{f}_t = \sigma(\mathbf{W}_f[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_f) \quad (15)$$

Input gate:

$$\mathbf{i}_t = \sigma(\mathbf{W}_i[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_i) \quad (16)$$

$$\mathbf{c}_t = \tanh(\mathbf{W}_c[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_c) \quad (17)$$

Cell state update:

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t \quad (18)$$

Output gate:

$$\mathbf{o}_t = \sigma(\mathbf{W}_o[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_o) \quad (19)$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t) \quad (20)$$

Where:

- $\mathbf{x}_t$ is the input vector at time t .
- $\mathbf{h}_{t-1}$ is the previous hidden state.
- $\mathbf{c}_{t-1}$ is the previous cell state.

- $\mathbf{f}_t, \mathbf{i}_t, \mathbf{o}_t$ are respectively the forget, input and output gate activations.
- $\mathbf{e}_t$ is the candidate cell state.
- $\mathbf{W}_*, \mathbf{b}_*$ are the weights and biases matrices.
- $\sigma()$ is the sigmoid activation function.
- $\tanh()$ is the hyperbolic tangent activation function.
- $\odot$ is the element-wise (Hadamard) product.

Algorithm 1 LSTM Cell Computational Steps

Require: Input vector x_t for the current time step, hidden state h_{t-1} from the previous step, cell state c_{t-1} from the previous step.

Ensure: Updated hidden state h_t for the current step, updated cell state c_t for the current step.

- 1: Determine the forget gate's activation f_t : $f_t \leftarrow \sigma(W_f[h_{t-1}, x_t] + b_f)$
 - 2: Establish the input gate's activation i_t : $i_t \leftarrow \sigma(W_i[h_{t-1}, x_t] + b_i)$
 - 3: Formulate the candidate cell state values $\tilde{c}_t$: $\tilde{c}_t \leftarrow \tanh(W_c[h_{t-1}, x_t] + b_c)$
 - 4: The new cell state c_t is obtained by combining the prior state and candidate values:

$$c_t \leftarrow f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$$
 - 5: Calculate the output gate's activation o_t : $o_t \leftarrow \sigma(W_o[h_{t-1}, x_t] + b_o)$
 - 6: The updated hidden state h_t is then derived: $h_t \leftarrow o_t \odot \tanh(c_t)$
 - 7: **return** h_t, c_t
-

3.3.5 N-BEATS (Neural Basis Expansion Analysis for Time Series)

N-BEATS is a deep neural network architecture designed specifically for time-series forecasting, which relies entirely on fully connected (feedforward) neural networks structured in blocks.

Each block learns to generate forecasts by combining basis expansions weighted by neural network outputs. It recursively refines the forecast by stacking multiple blocks, with each block focusing on improving the residuals left over from the previous block.

For each block b , the model computes the block features:

$$\theta^{(b)} = f_{NN}^{(b)}(\mathbf{x}^{(b)}) \quad (21)$$

Where $f_{NN}^{(b)}$ is the feedforward network at block b .

The second step is to calculate the backcast and forecast components:

$$\mathbf{b}^{(b)} = \mathbf{G}_b \theta^{(b)}, \mathbf{f}^{(b)} = \mathbf{H}_b \theta^{(b)} \quad (22)$$

where $\mathbf{G}_b$ and $\mathbf{H}_b$ are learned basis projection matrices.

Finally the residuals get updated

$$\mathbf{x}^{(b+1)} = \mathbf{x}^{(b)} - \mathbf{b}^{(b)} \quad (23)$$

With the final forecast being:

$$\hat{y} = \sum_{b=1}^B \mathbf{f}^{(b)} \quad (24)$$

Algorithm 2 N-BEATS Forecasting

Require: Input time series $y_{1:T}$, forecast horizon H

Ensure: Forecast $\hat{y}_{T+1:T+H}$

- 1: Initialize residual $r \leftarrow y_{1:T}$
 - 2: Initialize forecast $\hat{y} \leftarrow 0$
 - 3: **for** each stack $s = 1$ to S **do**
 - 4: **for** each block $b = 1$ to B in stack s **do**
 - 5: Compute block output: $[b_{\text{back}}, b_{\text{fore}}] \leftarrow f_b(r)$
 - 6: Update residual: $r \leftarrow r - b_{\text{back}}$
 - 7: Update forecast: $\hat{y} \leftarrow \hat{y} + b_{\text{fore}}$
 - 8: **end for**
 - 9: **end for**
 - 10: **return** Final forecast $\hat{y}$
-

3.3.6 Naive-12

The Naive-12 method is a variant of the naive forecast method. It predicts future values by simply using the value from the same time point 12 steps earlier:

$$\hat{y}_t = y_{t-12} \quad (25)$$

This model is used for benchmarking purposes. If a more sophisticated model cannot outperform Naive-12, it generally suggests that the added complexity is not justified.

3.3.7 Vanilla Transformer

This model architecture was originally developed in 2017 for natural language processing (NLP) and introduced a new paradigm for modeling data without relying on recurrence or convolutions (Vaswani et al., 2023). It uses a self-attention mechanism to capture dependencies between all elements in a sequence, regardless of their distance.

Each token in the sequence $(x_1, x_2, \dots, x_n)$ is mapped into a vector (Vaswani et al., 2023):

$$\mathbf{z}_i^0 = E(x_i) + \mathbf{p}_i \quad (26)$$

where:

- $E(x_i)$ is the token embedding.
- $\mathbf{p}_i$ is the positional encoding to preserve the sequence order.

Because the architecture itself does not include recurrence or convolutions to keep track of token positions, *positional encodings* are injected into the input data:

$$p_{i,2k} = \sin\left(\frac{i}{1000^{2k/d}}\right), p_{i,2k+1} = \cos\left(\frac{i}{1000^{2k/d}}\right) \quad (27)$$

The attention allows each position to access all other positions:

$$Attention(Q, K, V) = softmax \frac{QK^T}{\sqrt{d_k}} V \quad (28)$$

This allows the model to weigh the relevance of each token to every other, no matter how far apart they are. To capture multiple types of relationships, the Transformer uses multiple attention heads. Where each head is a separate attention computation with its own projections.

$$MultiHead(Q, K, V) = Concat(head_1, ..., head_h)W^O \quad (29)$$

After the attention layer, each position passes through a small feedforward network (Vaswani et al., 2023). This gives the model the ability to model nonlinearity and projection power independently at each position.

$$FFN(z) = max(0, zW_1 + b_1)W_2 + b_2 \quad (30)$$

In sequence-to-sequence tasks (like translation), the architecture has an encoder for processing the input sequence, and a decoder that generates the output in an autoregressive manner. It is important to note that the decoder uses masked self-attention so that when predicting position t it cannot peek ahead at position $t + 1$.

In time series forecasting in particular, the transformer processes a window of past observations and predicts future values. The inputs are continuous observations:

$$\mathbf{x}_t = [y_{t-p}, y_{t-p+1}, ..., y_t] \quad (31)$$

These include:

- Past targets y_t .
- Exogenous covariates like carbohydrate intake, bolus and basal insulin.

- Time features like hour of the day, day of the week.

These inputs are embedded using either learned or sinusoidal positional encodings.

$$\mathbf{z}_i^0 = E(x_i) + \mathbf{p}_i \quad (32)$$

The core attention mechanism still remains the same, however, in time series forecasting causal masking is used to ensure no future values are seen during training

- Encoder only looks at the past window.
- if the decoder is used it attends autoregressive, feeding it own prediction back in.

$$Attention(Q, K, V) = softmax \frac{QK^T}{\sqrt{d_k}} V \quad (33)$$

The goal is to predict the time series, which can be done in 2 different ways:

- The transformer can do direct forecasting using a multi-output head.
- Autoregressive (one step at a time).

$$\hat{y}_{t+1}, \hat{y}_{t+2}, \dots, \hat{y}_{t+h} \quad (34)$$

The final prediction head is usually a linear layer:

$$\hat{y}_{t+h} = \mathbf{W}^{out} \mathbf{z}_{t+h} + \mathbf{b}^{out} \quad (35)$$

For non probabilistic models like quintile or distributional forecasts the loss function is:

$$L(\theta) = \frac{1}{H} \sum_{h=1}^H (y_{t+h} - \hat{y}_{t+h})^2 \quad (36)$$

Algorithm 3 Transformer for Time Series Forecasting

Require: Input time series $X = [x_1, x_2, \dots, x_T]$, forecast horizon H

Ensure: Forecast $\hat{Y} = [\hat{y}_{T+1}, \dots, \hat{y}_{T+H}]$

- 1: Embed input: $E_X \leftarrow \text{Embed}(X) + \text{Positional Encoding}(T)$
 - 2: Pass through encoder: $Z \leftarrow \text{Encoder}(E_X)$
 - 3: Initialize decoder input $D \leftarrow \text{Embed}([0, \dots, 0]) + \text{Positional Encoding}(H)$
 - 4: **for** each decoding step $h = 1$ to H **do**
 - 5: Compute self-attention in decoder
 - 6: Compute encoder-decoder attention between Z and decoder
 - 7: Update decoder output D_h
 - 8: **end for**
 - 9: Generate forecast: $\hat{Y} \leftarrow \text{Output Projection}(D)$
 - 10: **return** $\hat{Y}$
-

3.3.8 Temporal Fusion Transformer

Temporal Fusion Transformers (TFT) is a specialized deep learning architecture for time series forecasting that builds upon the transformer architecture. It combines the attention mechanisms with recurrent layers and gating components, making it an ideal candidate for capturing both short-term and long-term temporal dependencies while also offering interpretability through attention and variable selection (Lim et al., 2020).

The Temporal Fusion Transformer can handle three types of covariates (Lim et al., 2020).

- static covariates $\mathbf{s}$.
- past observed covariates $\mathbf{y}_{1:t}$.
- know future covariates $\mathbf{x}_{t+1:t+H}$.

All the inputs are embedded using learned embedding matrices into a shared latent space (Lim et al., 2020):

$$\mathbf{e}_j = E_j \mathbf{x}_j \quad (37)$$

Where E_j is the embedding matrix for variable j . The variable selection network applies

a soft gating mechanism:

$$\mathbf{v}_j = \sigma(W_j^{sel}e_j + b_j^{sel}) \quad (38)$$

Where σ is the sigmoid activation function, and $W_j^{sel}e_j, b_j^{sel}$ are learned weights. This returns a weighted combination of inputs, allowing the model to focus on the most important variables.

The Gated Residual Network (GRN) processes non-linear inputs and adds gated residual connections (Lim et al., 2020):

$$\mathbf{g}(z) = LayerNorm((ELU(zW_1 + b_1)W_2 + b_2) + z) \quad (39)$$

This formula ensures stability and more flexible transformations, with residual and normalization components that aid in the training process. To handle temporal pattern TFT uses a combination of a recurrent layer (LSTM) and self-attention layer.

The recurrent layer is:

$$h_t = LSTM(h_{t-1}, z_t) \quad (40)$$

Where h_t is the hidden state capturing local sequential information. The Self-Attention layer:

$$Attention(Q, K, V) = softmax(\frac{QK^T}{\sqrt{d_k}})V \quad (41)$$

Where Q, V, K are the query, value, and key matrices derived from hidden states. This layer aids the model in long-term dependency modeling by allowing each time step to attend to both past and future time points.

Static covariates s are processed by the GRNs and used to condition both the temporal layers and the final output layers by embedding information that does not change over time (e.g. patient id). When performing multi-horizon prediction for each forecast step h , the

model outputs an equation where z_{t+h} aggregates the LSTM outputs, the attention outputs, and the static covariates (Lim et al., 2020).

$$\hat{y}_{t+h} = W^{out} z_{t+h} + b^{out} \quad (42)$$

Unlike the traditional vanilla transformer the temporal fusion transformer does not do point estimate prediction, but by default it does quintile prediction, suing the quantile loss function (Lim et al., 2020):

$$L(\theta) = \frac{1}{H} \sum_{h=1}^H \sum_{q \in Q} \max(q(y_{t+h} - \hat{y}_{t+h}^q), (q-1)(y_{t+h} - \hat{y}_{t+h}^q)) \quad (43)$$

Where Q is the set of quantiles (e.g. 0.1, 0.5, 0.9), y_{t+h} is the true value, and $\hat{y}_{t+h}^q$ is the predicted quintile.

Algorithm 4 Temporal Fusion Transformer (TFT) for Time Series Forecasting

Require: Past inputs $X_{1:T}$, known future inputs $X_{T+1:T+H}$, static features S

Ensure: Forecast $\hat{Y}_{T+1:T+H}$

- 1: Apply variable selection network:
 - 2: Select relevant static features S' and temporal features X'
 - 3: Encode static features:
 - 4: Compute static context vectors (for conditioning)
 - 5: Process historical inputs:
 - 6: Pass $X'_{1:T}$ through LSTM encoder
 - 7: Process future inputs:
 - 8: Pass $X'_{T+1:T+H}$ through LSTM decoder (conditioned on encoder and static context)
 - 9: Apply multi-head attention:
 - 10: Attend over decoder outputs with temporal attention layer
 - 11: Apply gating mechanisms and skip connections:
 - 12: Combine attention outputs, context vectors, and LSTM outputs
 - 13: Pass through position-wise feed-forward network
 - 14: Generate final predictions:
 - 15: $\hat{Y}_{T+1:T+H} \leftarrow \text{OutputLayer}(\text{final representations})$
 - 16: **return** $\hat{Y}_{T+1:T+H}$
-

3.4 Model Evaluation

3.4.1 Performance Metrics

To evaluate the model performance, two complementary error metrics were used, each offering different insights into the accuracy and the reliability of the predictions.

- Root Mean Squared Error
- Mean Absolute Error

Root Mean Squared Error (RMSE) gives weight to larger errors because of the squaring operation, making it more sensitive to outliers. This sensitivity is particularly valuable when performing glucose prediction because large deviations can indicate dangerous hypoglycemic or hyperglycemic events that may require immediate intervention. A lower RMSE indicates that the model can produce predictions that are close to the actual values, especially in terms of avoiding large and potentially critical errors. However, because RMSE can be easily skewed by just a couple of large mistakes, it is often interpreted alongside other metrics that offer a more balanced assessment.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (44)$$

Mean Absolute Error (MAE) provides a balanced average of absolute errors, offering a more interpretable and robust way to measure the errors that is less affected by sparse but big prediction errors. Unlike RMSE, MAE takes into consideration all of the errors equally, making it an overall good general-purpose indicator of model accuracy. In practice, MAE can help quantify the average error distance that the prediction makes compared to the actual values. This provides a more intuitive way of understanding how far off predictions are on average from their true values.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (45)$$

By using both RMSE and MAE in conjunction with each other, it allows us to get a more comprehensive view of how the model is performing, balancing sensitivity to large deviations with a reliable measure of overall performance accuracy. This dual-metric ensures that critical safety errors and general prediction quality are both fairly assessed.

3.4.2 Clinical Relevance Assessment

When dealing with medical data it is not enough to just evaluate the statistical performance of a model. It is crucial to also assess the clinical significance of the prediction errors, since even some small inaccuracies in the prediction can have vastly different impacts on patient safety. For this reason, the Clarke Error Grid (CEG) Analysis was used, as it is a well-established method for evaluating the clinical accuracy of self-monitoring blood glucose (SMBG) devices, designed to assess how predictions translate into real-world clinical risks.

The way the Clarke Error Grid works is by comparing predicted glucose values against the actual glucose values and categorize each prediction into five distinct zones, which indicates their potential clinical risk.

- **Zone A** are clinically accurate predictions, the results are within 20% from the reference value or both in the hypoglycemic ($< 70 \frac{mg}{dL}$). Predictions in this zone are considered safe and would lead to correct treatment decisions (Clarke et al., 1987).
- **Zone B** are predictions outside of 20% reference point but are unlikely to result in inappropriate treatment or adverse clinical outcomes. While these predictions are not considered perfectly accurate, they are acceptable from a clinical management standpoint (Clarke et al., 1987).
- **Zone C** are over-correction errors; these predictions can lead to unnecessary or erroneous treatment, such as administering glucose when it is not needed, increasing the risk of causing glucose level swings (Clarke et al., 1987).

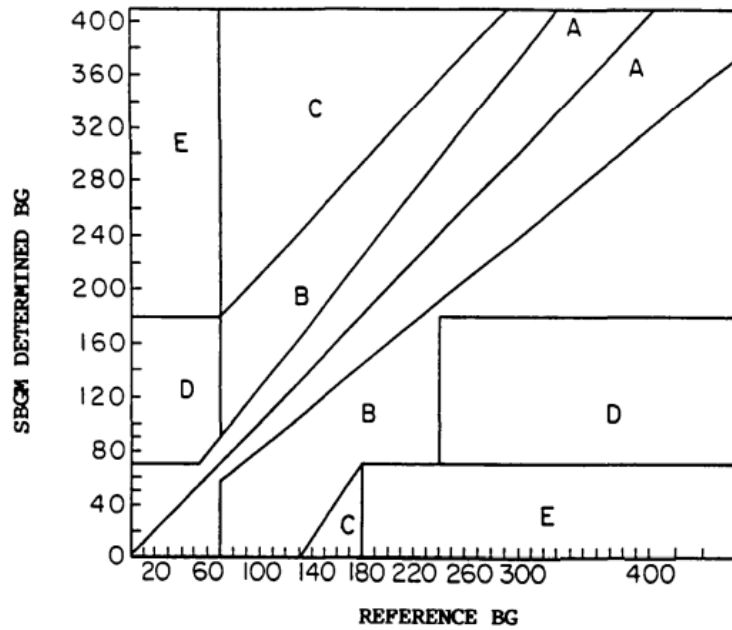


Figure 3: Clarke Error Grid (Clarke et al., 1987)

- **Zone D** are dangerous misses; these are failures to detect hypoglycemic or hyperglycemic events, where the model either underestimates or overestimates glucose levels, potentially leading to a lack of critical intervention and posing serious health risks (Clarke et al., 1987).
- **Zone E** is erroneous treatment; this is the most severe category, where predictions result in a complete misclassification (e.g., mistaking hypoglycemia for hyperglycemia or vice versa), leading to harmful opposite treatments that can be life-threatening (Clarke et al., 1987).

The x axis of the graph indicates the reference blood glucose level, and the y axis indicates the predicted blood glucose level.

The proportion of predictions falling into Zone A and Zone B are indicative of models that are both accurate and clinically safe, as they either match their corresponding reference values or closely resemble them, introducing only minor and non-dangerous discrepancies. The lower percentage zones C, D, and E highlight some potential clinical risk, where the model prediction could result in harmful or inappropriate medical actions.

By incorporating Clarke Error Grid analysis alongside statistical error metrics, it helps obtain deeper insights into the the model's real-world performance, ensuring that the prediction accuracy is not just statistical but also aligns with clinical safety and decision-making requirements.

3.4.3 Backtesting

To ensure a realistic assessment of model performance, this study first performed backtesting on the complete 28-patient dataset. Backtesting focuses on evaluating the model's ability to make predictions using only past information, which mimics the real-time deployment conditions.

In this approach, each patient time series was split into two parts: 90% was for training and the remaining 10% was for backtesting. The models were trained on the historical portion of the data, and then used to generate predictions 12 steps at a time. This method allows for unbiased validation of how a model generalizes to unseen data, ensuring no artificial inflation in performance metrics.

Backtesting was conducted on all 28 patients, providing a robust evaluation of both temporal accuracy and variability across patients. This was the first step used as a screening phase to identify the most promising models within the linear and non-linear categories. Based on the backtesting results, the top-performing models were selected for a more rigorous evaluation using the expanding window cross-validation method.

This technique helps reveal if the model can only fit well to past patterns or if it can be genuinely applied to unseen future data.

3.4.4 Cross-Validation

To conduct a robust evaluation of the model on time series data, this study took the best-performing models in the linear and non-linear category and employed cross-validation using an expanding window approach, across the data collected from 28 patients. Un-

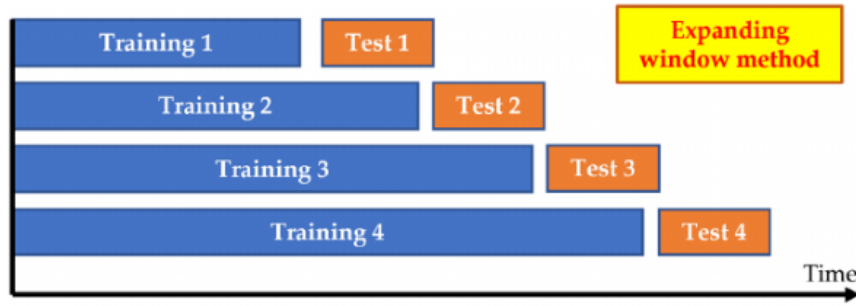


Figure 4: Expanding Window (Bera et al., 2023)

like traditional cross-validation methods that randomly divide data into training and testing sets, time series cross-validation must preserve the chronological sequence of the data. This is crucial because using future data to predict the past introduces data leakage and overestimates the model performance.

In this approach, the training window began with an initial size, and as the folds increased so did the windows, progressively incorporating more historical data. This method allows the model to continuously learn from a growing base of past information, mirroring real-world clinical scenarios where more patient data becomes available the longer they wear the device.

Across all folds the model was retrained on the updated training window and evaluated on the corresponding test window, providing performance estimates at multiple points in the time series.

Although data were available from 28 patients, 3 patients (1 adult, 1 adolescent, 1 child) were deliberately excluded from the cross-validation. This was done to make sure that the model did not overfit to specific patient data but could generalize well to new unseen patients as well.

By averaging performance metrics across all folds and patients, the expanding window approach provides a comprehensive and realistic estimate of model generalizability and stability. This technique gives a perspective on how the model adapts over time and also how robust it is across different individual patients. This method is particularly well-suited

for applications like glucose prediction, where both temporal dynamics and inter-patient variability must be taken into consideration to ensure a clinically safe model.

3.5 Hardware and Software Implementation

All the models were implemented using `Python 3.10.12` with the following key libraries:

- `pandas` and `Numpy` for data manipulation.
- `matplotlib` for data visualization.
- `scikit-learn` data pre-processing, and scaling.
- `darts` for creating all of the models, both deep learning and statistical models.
- `Optuna` was used to perform hyper-parameter tuning.
- `TensorBoard` was used for experiment tracking during hyper-parameter tuning, and during final training.

All of the models were trained and evaluated on a laptop equipped with:

- NVIDIA GeForce RTX 4060 Laptop GPU with 8Gb of VRAM
- 13th Gen Intel(R) Core(TM) i9-13900H
- 32Gb of RAM

4 Results

This section presents the detailed outcomes of the experiments conducted, evaluating multiple forecasting models on 28 simulated diabetic patients generated using the UVA/PADOVA simulator. Each patient dataset contained 8640 time steps, corresponding to 30 continuous days of 5-minute interval continuous glucose monitoring (CGM). A 90-10 train-test

split was performed, reserving 864 data points (approximately 3 days) for backtesting. The prediction task focused solely on forecasting 12 time steps ahead (1 hour), using a stride of 12.

4.1 Overall Performance Metrics

The models that were implemented and evaluated are as follows:

- **Linear Models:**

- ARIMA
- Exponential Smoothing
- Linear Regression
- Naive 12

- **Non-linear Models:**

- LSTM
- N-BEATS
- TFT
- TFT (Tuned)
- Vanilla Transformer

The comparison metrics included mean absolute error (MAE), root mean squared error (RMSE), inference time, and Clarke's Error Grid (CEG) zones A–E.

4.1.1 Box Plots

Figure 5 and Figure 6 show the MAE and RMSE box plots, respectively. Each box plot summarizes the spread and variability of MAE and RMSE across 28 patients. The central

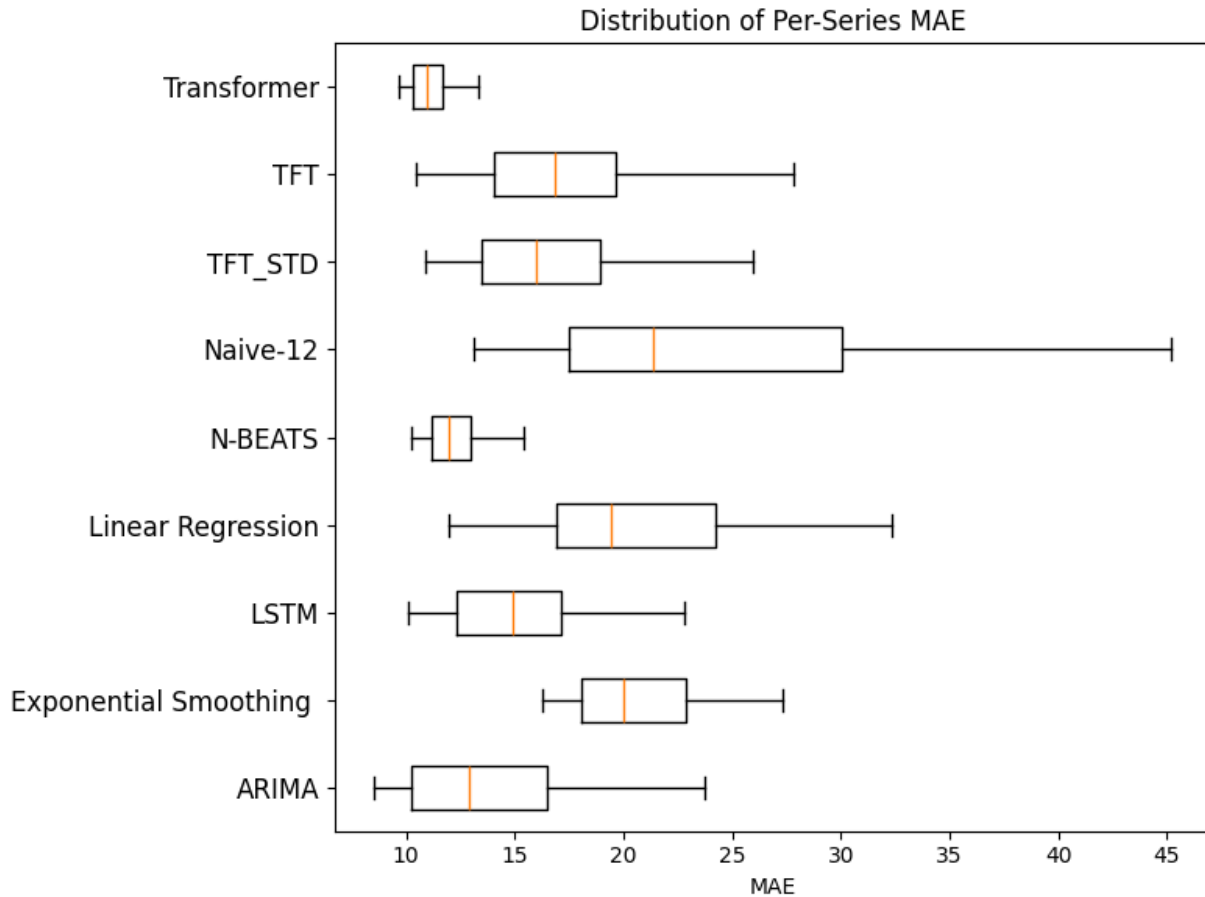


Figure 5: Mean Absolute Error Box Plot

orange line indicates the median, while the box boundaries represent the interquartile range (IQR), capturing the middle 50% of the errors. The whiskers extend to the minimum and maximum observed errors across patients.

It is evident that non-linear models such as the Vanilla Transformer and N-BEATS exhibit more tightly packed boxes and shorter whiskers, indicating consistent performance between patients. In contrast, models like Naive-12 and Linear Regression show wider boxes and longer whiskers, suggesting greater variability and occasional poor predictions.

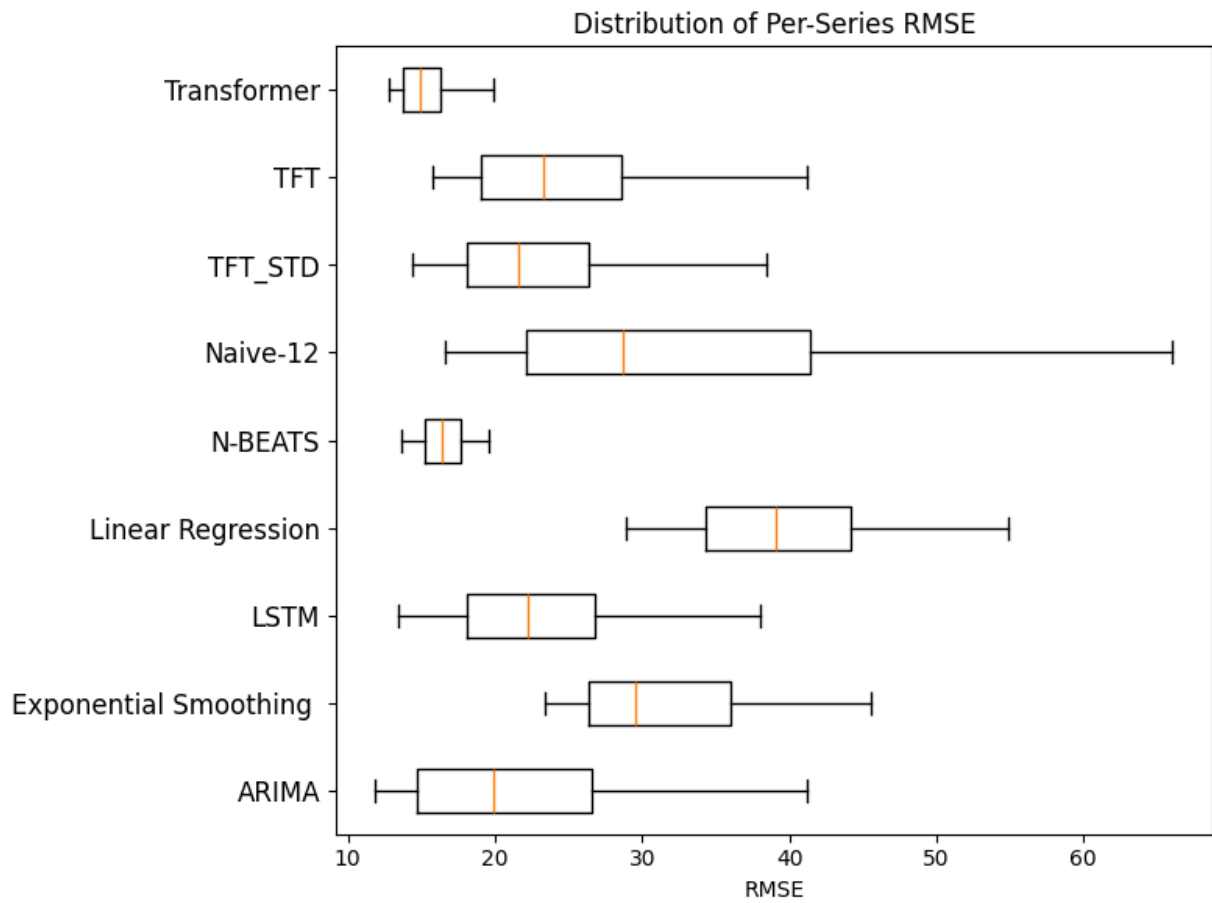


Figure 6: Root Mean Squared Error Box Plot

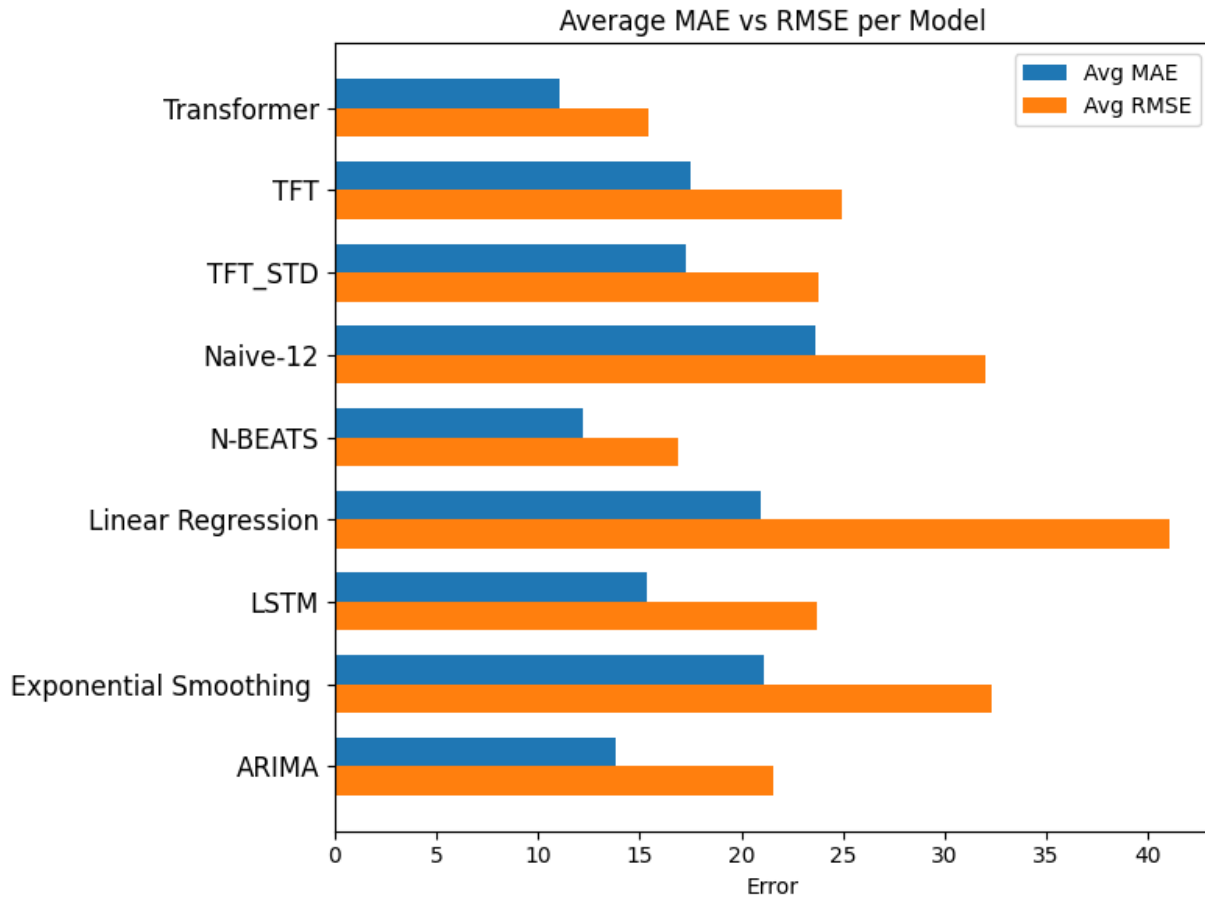


Figure 7: MAE and RMSE Histogram

4.1.2 Histogram

The histogram in Figure 7 compares model performance by plotting the median RMSE and median MAE. This plot focuses solely on predictive accuracy, ignoring computational cost. The Transformer and ARIMA models stand out in their respective categories, achieving low median MAE and RMSE and confirming their reliability.

4.1.3 Time-Based Plot

Figures 8 and 9 plot the median MAE and RMSE against inference time for each model. The x-axis shows elapsed time in seconds (log scale), while the y-axis shows the error

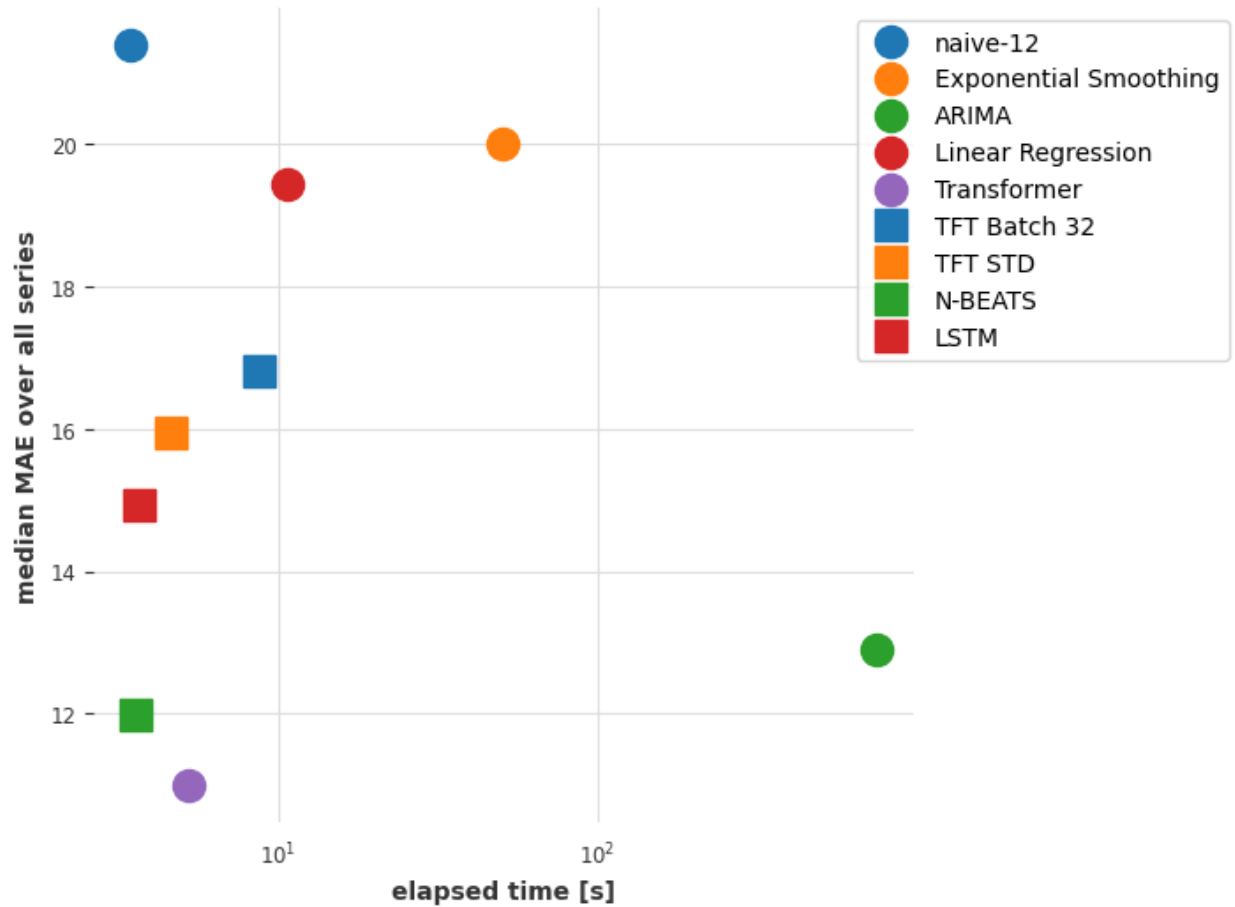


Figure 8: Mean Absolute Error vs. Inference Time

metric. These plots reveal the trade-off between predictive performance and computational efficiency: some models achieve low error but incur high computational cost.

4.1.4 Temporal Fusion Transformer Performance and Hyperparameter Tuning

The Temporal Fusion Transformer (TFT) underperformed compared to simpler models like the Vanilla Transformer and ARIMA. To improve performance, hyperparameter tuning was performed using the `Optuna` library.

The following hyperparameters were tuned:

- `hidden_size`: 32–128

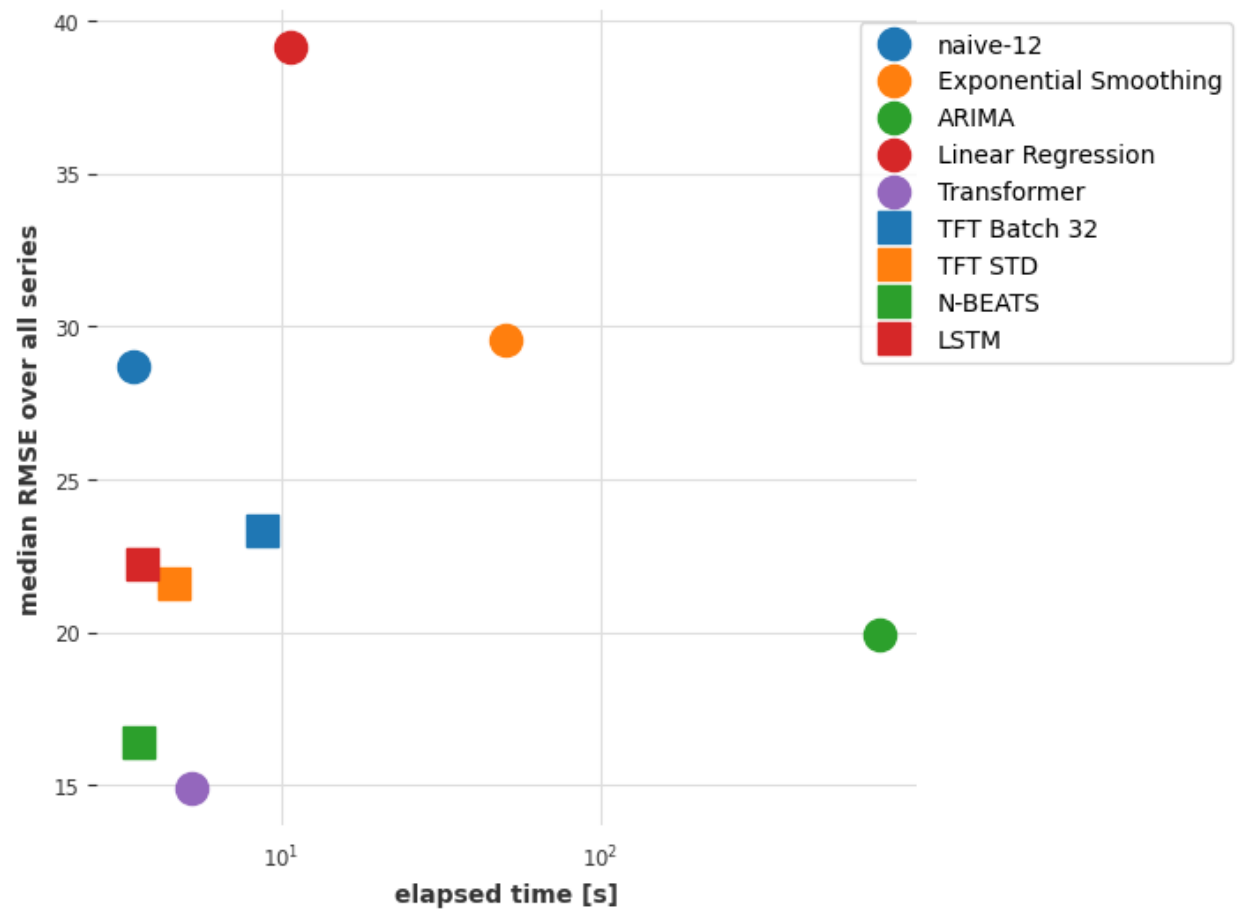


Figure 9: Root Mean Squared Error vs. Inference Time

- lstm_layers: 1–2
- num_attention_heads: 1–2
- batch_size: 32, 128, 512, 1024
- learning_rate: 1×10^{-4} to 1×10^{-2}

Early stopping was applied to avoid overfitting, with:

- patience = 3
- min_delta = 0.001

Figure 10 shows validation loss across 50 trials. Trial 5 yielded the best hyperparameters:

- hidden_size: 119
- lstm_layers: 1
- num_attention_heads: 2
- batch_size: 512
- learning_rate: 0.0002684671988513788

Despite tuning, improvements were marginal. Backtesting revealed that both tuned and untuned TFT overfit to CGM noise, limiting generalization.

Figure 11 shows the backtesting of the fine-tuned TFT.

4.1.5 Performance Summary Table

Table 1 summarizes model performance across all 28 patients.

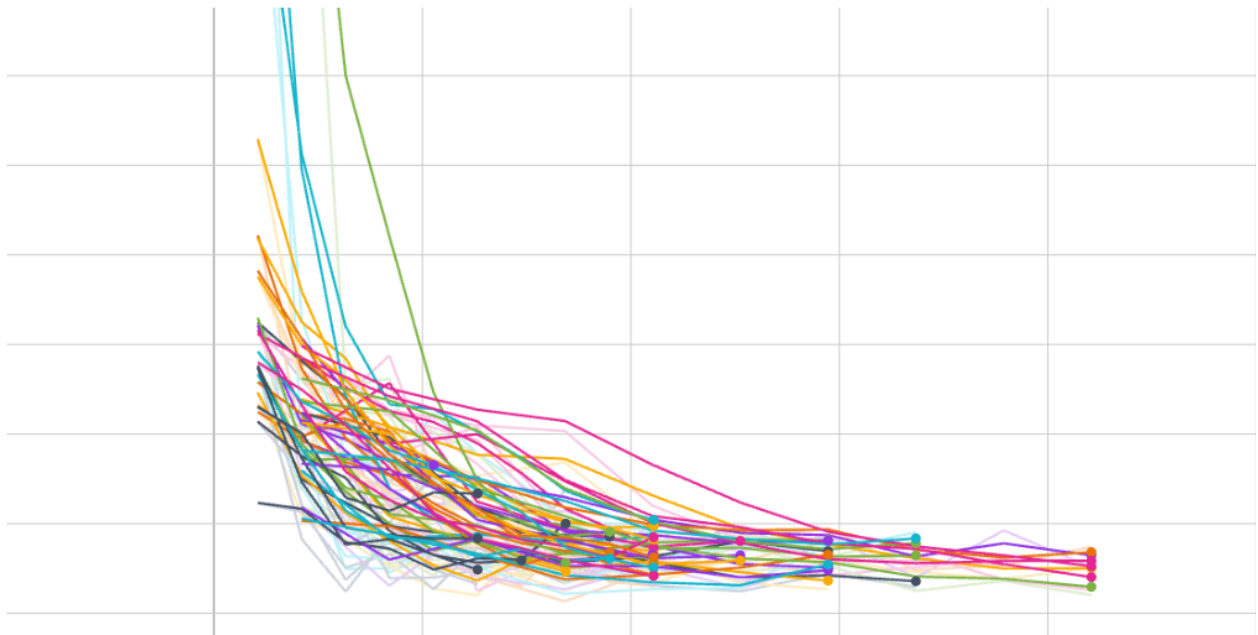


Figure 10: Validation Loss During Hyperparameter Tuning for TFT

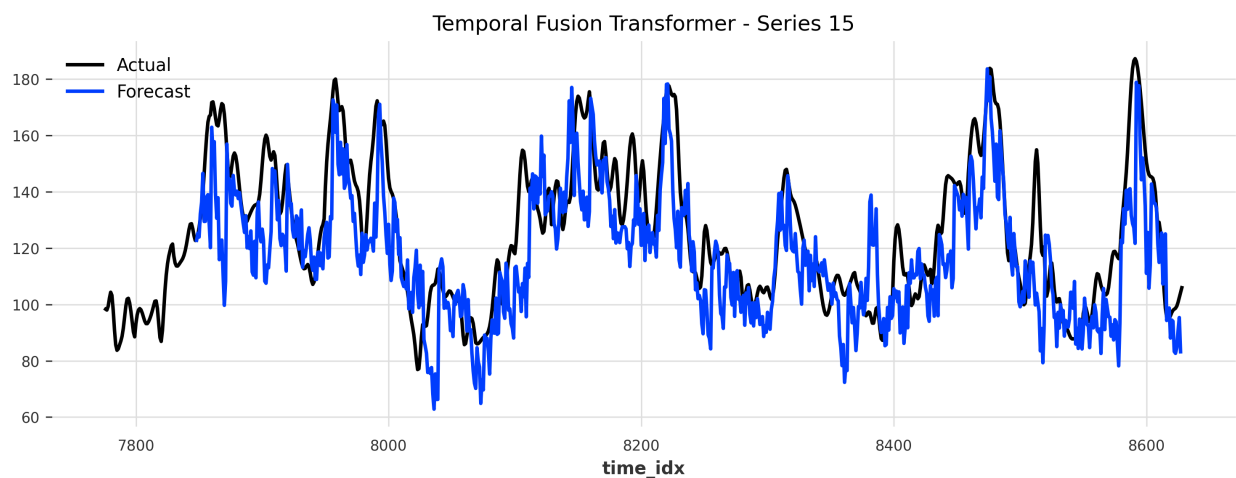


Figure 11: Fine-Tuned TFT CGM Backtesting on Validation Set

Table 1: Performance of Linear vs. Nonlinear Models (best in **bold**, second-best underlined)

Linear Models								
Model	MAE	RMSE	Time (s)	Zone A	Zone B	Zone C	Zone D	Zone E
ARIMA	13.80	21.56	749.91	83.09%	15.22%	0.00%	1.66%	0.03%
Exponential Smoothing	21.13	32.34	50.45	70.21%	26.77%	0.68%	2.11%	0.22%
Linear Regression	<u>20.95</u>	41.07	10.59	<u>74.22%</u>	20.95%	1.08%	3.55%	0.21%
Naive 12	23.61	<u>32.03</u>	3.44	66.01%	30.11%	0.30%	3.53%	0.04%
Nonlinear Models								
Model	MAE	RMSE	Time (s)	Zone A	Zone B	Zone C	Zone D	Zone E
LSTM	15.33	23.71	3.64	81.43%	15.40%	0.00%	3.14%	0.03%
N-BEATS	<u>12.17</u>	<u>16.87</u>	3.56	<u>88.57%</u>	10.72%	0.00%	0.71%	0.00%
TFT (Tuned)	17.24	23.79	4.57	77.11%	20.78%	0.00%	2.11%	0.00%
TFT	17.51	24.92	8.67	76.18%	22.21%	0.00%	1.61%	0.00%
Vanilla Transformer	11.07	15.44	5.21	90.19%	8.36%	0.03%	1.42%	0.00%

4.2 Cross-Validation and Backtesting

After evaluating all the models, the best performing linear model (ARIMA) and the best performing non-linear model (Transformer) were selected for further validation. To better asses their generalizability, a 10-fold cross validation using expanding window was performed. The cross-validation was performed on 25 out of the 28 patients patients, leaving 3 patiens out for validation.

Table 2: Cross-Validation Performance

Model	MAE	RMSE	Zone A	Zone B	Zone C	Zone D	Zone E
ARIMA	12.59	15.64	83.87%	13.77%	0.00%	2.37%	0.00%
Vanilla Transformer	12.47	14.45	87.23%	10.83%	0.00%	1.93%	0.00%

After performing the cross validation, backtesting was performed on the Vanilla Transformer to asses its performance for patients it has never seen.

Table 3: Transformer backtesting on the 3 left-out patients

Model	MAE	RMSE	Zone A	Zone B	Zone C	Zone D	Zone E
Vanilla Transformer	10.18	14.13	89.22%	8.88%	0.00%	1.90%	0.00%

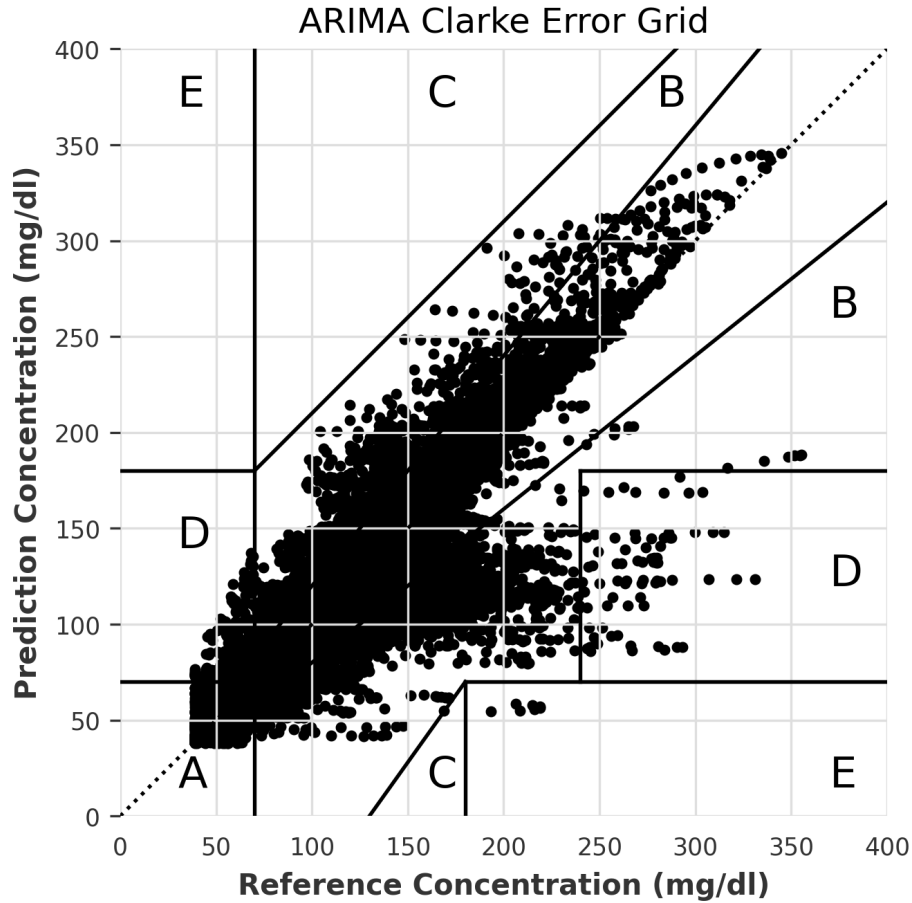


Figure 12: ARIMA Clarke's Error Grid

4.3 Clarke's Error Grid Analysis

Clinical relevance was assessed using Clarke's Error Grid, evaluating whether predictions fall into clinically acceptable zones.

Figures 13, 12, and 14 show the CEG scatter plots for the Vanilla Transformer, ARIMA, and Naive-12 models, respectively.

Both the Vanilla Transformer and ARIMA achieved over 95% predictions in Zones A and B, meeting clinical safety thresholds. In contrast, simpler models like Naive-12 showed more predictions in Zones C and D, indicating lower clinical reliability.

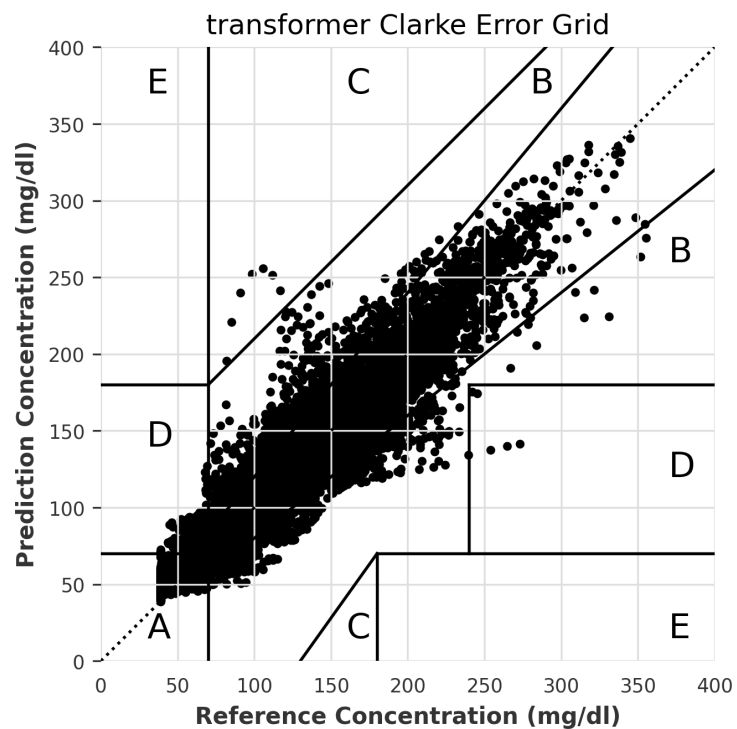


Figure 13: Vanilla Transformer Clarke's Error Grid

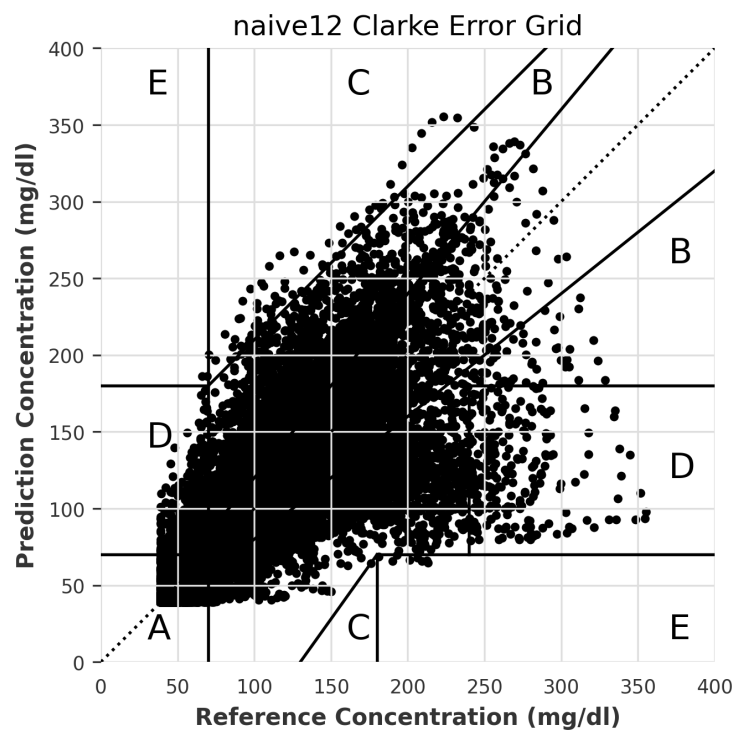


Figure 14: Naive 12 Clarke's Error Grid

5 Conclusion and Future Work

5.1 Conclusion

This thesis embarked on an extensive and rigorous comparative analysis of diverse time series forecasting models for short-term (60 minutes) glucose prediction in type 1 diabetes using a simulated dataset of 28 virtual patients with the use of the UVA/Padova Type 1 diabetes simulator. The primary objective was to evaluate the effectiveness of classical statistical methods, machine learning approaches, and sophisticated deep learning approaches. The overarching goal was to identify models that not only have high prediction scores but also demonstrate clinical safety and robust generalization capabilities.

The findings of this evaluation provided some key insights into the current landscape of glucose forecasting. The traditional transformer architecture distinguished itself as the most potent non-linear architecture. It consistently had the lowest MAE score ($11.07 \frac{mg}{dL}$) and RMSE ($14.45 \frac{mg}{dL}$) during the initial backtesting phase across all 28 simulated patients. To prove that this robustness was not due to a single data split, the model was further re-trained through a demanding 10-fold expanding window cross-validation on 25 patients. Crucially, when tested on 3 entirely unseen patients of different age groups, the transformer managed to keep its excellent performance (MAE: $10.18 \frac{mg}{dL}$, RMSE: $14.13 \frac{mg}{dL}$). From a clinical safety perspective, the transformer managed to hit 98.1% of its predictions falling within the clinical Zone A (89.22%) and Zone B (8.88%), indicating a very low risk of, and a low probability that, predictions could lead to harmful treatment decisions.

Among the linear models, the ARIMA model was the most effective. While much simpler than its deep learning counterparts, ARIMA managed to provide a very strong baseline with a MAE of $13.80 \frac{mg}{dL}$ and an RMSE of $21.56 \frac{mg}{dL}$ in the initial backtesting phase. In the cross-validation phase, ARIMA managed to keep good prediction scores (MAE: $12.59 \frac{mg}{dL}$, RMSE: $15.64 \frac{mg}{dL}$) with a clinical safety profile of 97.64%, but at a very high computational cost during the backtesting phase because it needed to be re-trained every time (12.4

minutes).

In contrast, the Temporal Fusion Transformer (TFT), an architecture specifically designed from the ground up for multi-horizon time series forecasting, did not live up to its theoretical promise in this particular application. Despite its sophisticated design with recurrent layers, gating mechanisms, and attention, and even a dedicated hyperparameter tuning attempt, TFT was outperformed by both the vanilla transformer architecture and the N-BEATS architecture. A critical observation was that the TFT had a tendency to overfit to the noisy CGM signal, which limited its ability to generalize effectively. This finding suggests that while architectural complexity can be beneficial, it can also become a detriment. More simple models like exponential smoothing, linear regression, and naive-12, while computationally very efficient and easy to implement, gave significantly worse predictions than other more sophisticated models. A cornerstone of this research was the emphasis on a robust evaluation framework, going beyond standard statistical metrics (MAE, RMSE), which can sometimes be misleading if not contextualized, by incorporating Clarke Error Grid analysis, which was indispensable to assess the clinical implications of the prediction errors. Furthermore, the validation methodology combined an initial 90-10 split, and then performed backtesting on all patients. Secondly, for the best performing models in their class, an expanding window cross-validation test was done, followed by a final backtesting on completely held-out patients, providing a realistic and trustworthy assessment of model performance, generalizability, and potential for overfitting. While deep learning models like the Vanilla Transformer and N-BEATS offered superior accuracy, they also came with a degree of “black box” opacity compared to more simple models like ARIMA. The use of the UVA/Padova simulator was a deliberate choice, offering distinct advantages over real-world data, such as the absence of missing data and perfect meal and insulin logging. This allowed for a fair and focused comparison between architectures without confounding factors often present in messy real-world clinical datasets. However, this strength is also a limitation because no matter how sophisticated and advanced a

simulation is, it can never perfectly replicate the full spectrum of unmodeled physiological disturbances, sensor artifacts, and behavioral inconsistencies encountered in real life.

In summary, this thesis makes a significant contribution by systematically evaluating a broad range of forecasting models for glucose prediction under a robust and clinically relevant framework. The findings clearly show the forecasting potential of deep learning models. The results provide a valuable benchmark and evidence base that can aid in future development and refinement of technologies aimed at improving the quality of life of patients with type 1 diabetes and also reduce the burden on the healthcare system.

5.2 Limitations of the study

While this thesis provides meaningful insights, it is important to recognize its limitations and to contextualize the findings appropriately to guide future investigation.

The main constraint lies in the reliance on the UVA/Padova Type 1 Diabetes Simulator. Although FDA-approved and highly sophisticated, simulated data does not fully replicate the complexities and variability found in real-world scenarios. The data does not include unforeseen physiological variability. Some examples are: acute illness, severe stress, and changes in medication unrelated to diabetes, which are not present in the data. Another limiting factor is sensor noise and specific artifacts. While the simulation includes some noise, it only made use of one specific CGM sensor. Different sensors can exhibit different noise patterns, signals, calibration issues, and drift over the sensor's lifespan. In addition, meal logging in the real world can be inaccurate, and insulin dosing can also be missed or miscalculated, while the simulation always assumes the perfect case scenario. Moreover, while the simulation is capable of simulating adults, adolescents, and children, real-world populations exhibit greater variance in terms of insulin sensitivity, disease duration, and lifestyles. Additionally, this specific thesis focused exclusively on a 60-minute prediction horizon. While clinically relevant for many interventions, some other applications like rapid hypoglycemia alerts might require a shorter prediction, or for meal planning or overnight

control applications, it requires longer prediction horizons. Another limiting factor is that the best performing models, like Transformer and N-BEATS (though N-BEATS has some interpretable components), operate essentially as “black boxes”, which means that there is a lack of interpretability, making clinicians and patients more hesitant to trust a model without fully understanding the underlying logic behind the predictions.

5.3 Future Work

The insights and limitations pave the way for several promising avenues of future research, aimed at enhancing the accuracy of prediction, increasing the reliability of the models, and enhancing the practical utility of glucose prediction systems.

5.3.1 Validation and Adaptation on Diverse Real-World Clinical Datasets

The first step is to rigorously validate the top-performing models on a large-scale, multi-center real-world clinical dataset. These datasets should include a diverse set of patients with different lifestyles, CGM devices, and varying levels of data quality. This will test the true generalizability and robustness of the models compared to the complex and error-prone real-life scenarios. This will allow for a more realistic assessment of their clinical potential, alongside highlighting potential problems that models might need fine-tuning to better handle real-life data.

5.3.2 Exploration of Dynamic and Longer Prediction Horizons

The next step would be to investigate the model performance across a different range of prediction horizons (e.g., 30, 90, 120 minutes). Longer horizons are beneficial for more strategic planning, but they are harder to accurately predict due to error accumulation.

5.3.3 Advanced Personalization and Transfer Learning Strategies

While the current model provides generalizable predictions, integrating transfer learning through the use of patient-specific fine-tuning of a pre-trained population model could improve the per-patient predictions due to the high variability of T1D.

5.3.4 Incorporation of a Richer Set of Contextual and Physiological Data

Integrating as covariates more detailed macronutrient information such as: sleep quality metrics, physiological stress markers like heart-rate variability, or even data on illness and menstrual cycles could provide more context to understand and predict glucose fluctuations more accurately.

5.3.5 Explainable AI (XAI) for Glucose Prediction

One big downside of “black box” models is their interpretability. There are several XAI techniques such as SHAP, LIME, and attention visualization that allow us to gain insights into how the models arrived at a specific prediction. Enhancing model interpretability can increase trust among clinicians and patients, as well as help in debugging model performance.

By systematically addressing these future research directions, the scientific and medical community can continue to innovate in glucose forecasting. The ultimate aspiration is to develop a high-accuracy, reliable, personalized, and interpretable prediction model able to alleviate the burden of T1D in patients. This thesis represents a step on that important journey, offering a solid foundation and some clear signposts for the path ahead.

6 Code Availability

The complete source code used for the model implementations, data preprocessing, analysis, and generation of figures in this thesis is publicly available on GitHub <https://github.com/SpyrosMitsis/time-series-glucose-prediction-t1d>. A static version of the code corresponding to this thesis has also been archived on Zenodo at <https://doi.org/10.5281/zenodo.15504215>.

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